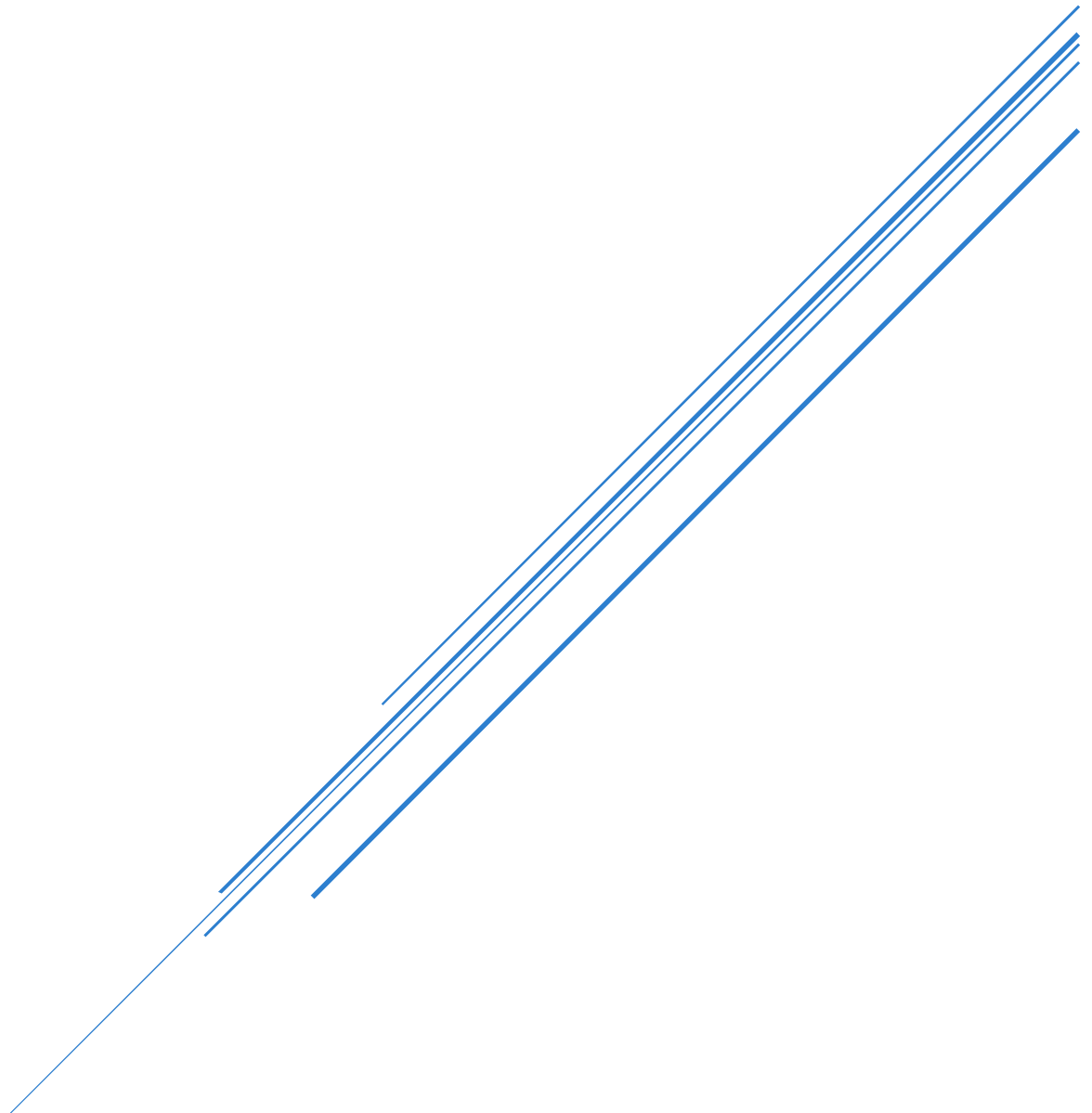


ONLINE SHOPPERS BEHAVIORAL ANALYSIS

An ML pipeline guidance



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1. Introduction

This project focuses on analysing online shopper behaviour using the Online Shoppers Purchasing Intention dataset. The primary objective is to predict whether a user is likely to make a purchase during a session and to segment users based on their behavioural patterns.

With the rapid growth of e-commerce, customers increasingly rely on online platforms for browsing, comparing, and purchasing products. Even when purchases are made in physical stores, online platforms are often used to check product availability and pricing. As a result, understanding user behavior on e-commerce websites has become essential for businesses.

The dataset captures various aspects of user interaction, including browsing activity, session duration, traffic sources, and engagement metrics. These features provide valuable insights into customer preferences, website usage patterns, and potential purchasing behaviour.

Such insights can be leveraged to improve personalization, optimize marketing strategies, and enhance the overall customer experience. Businesses can identify high-value customers, detect potential drop-off points, and allocate resources more effectively.

Ultimately, this project aims to:

- Predict whether a user will complete a purchase
- Segment users based on behavioural patterns to support data-driven decision-making

2. Problem Statement

The primary challenge addressed in this project is not merely building a classification model but developing a complete and reusable machine learning pipeline that can support both training and real-time prediction.

While implementing basic classification models has become relatively straightforward, integrating preprocessing, model training, evaluation, and prediction into a structured pipeline remains a critical requirement for practical applications.

This project aims to design an end-to-end pipeline that:

- Processes raw behavioural data
- Trains and evaluates multiple machine learning models
- Generates predictions and probability scores
- Supports deployment through an interactive dashboard

The core task is formulated as a binary classification problem, where the goal is to predict whether a user session results in a purchase (Revenue = True) or not (Revenue = False). In addition to prediction, the project extends its value by segmenting users into behavioural categories from high-

intent buyers to low-intent visitors. This enables more actionable insights for business use cases, including targeted marketing, personalization, and conversion optimization.

3. Dataset Description

- Online Shoppers Purchasing Intention Dataset downloaded from Kaggle
- Number of rows – 12,330
- Types of features:
 - Numerical
 - Administrative
 - Administrative_Duration
 - Informational
 - Informational_Duration
 - ProductRelated
 - ProductRelated_Duration
 - BounceRates
 - ExitRates
 - PageValues
 - SpecialDay
 - Categorical
 - Month
 - VisitorType
 - Weekend
 - OperatingSystems
 - Browser
 - Region
 - TrafficType
 - Target variable: Revenue

4. Data Preprocessing

- Handling missing values – The dataset I used initially for the model training did not have any missing values. However, it is not always the case. The new datasets uploaded to the dashboard to get predictions may have missing values. Therefore, in the preprocessing stage, I have added sections to handle missing values.
 - Numerical values – with median
 - Categorical values – with “Unknown”
 - Boolean values – with “False”
- Encoding categorical variables – One-Hot Encoding is used in the preprocessing stage to convert categorical text data into binary matrix.
- Feature scaling – This is not included in the preprocessing stage of my pipeline as while LR model requires scaling, RF model does not need scaling.
- Column alignment for prediction

Preprocessing is designed to work for both training and testing.

5. Exploratory Data Analysis (EDA)

The EDA is carried out in “01_data_understanding.ipynb”.

Since it is crucial to comprehend the dataset in order to determine what kinds of models will work best, I conducted an EDA.

Let's start by examining our columns.

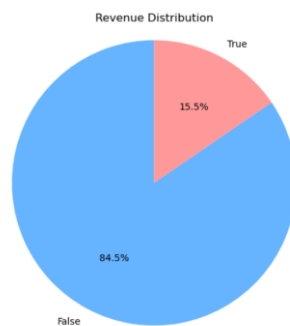
Numerical Features

- Administrative – Number of administrative pages visited during the session
- Administrative_Duration – Total time spent on administrative pages
- Informational – Number of informational pages visited
- Informational_Duration – Total time spent on informational pages
- ProductRelated – Number of product-related pages viewed
- ProductRelated_Duration – Total time spent on product-related pages
- BounceRates – Percentage of visitors who leave the site after viewing only one page
- ExitRates – Percentage of exits from the website after viewing a specific page
- PageValues – Average value of a page before a transaction occurs
- SpecialDay – Closeness of the session to a special day (e.g., holidays), ranging from 0 to 1

Categorical Features

- Month – Month in which the session occurred
- OperatingSystems – Operating system used by the visitor (encoded as category)
- Browser – Web browser used by the visitor (encoded as category)
- Region – Geographic region of the visitor (encoded as category)
- TrafficType – Source of website traffic (e.g., direct, referral, ads)
- VisitorType – Type of visitor (Returning or New)
- Weekend – Indicates whether the session occurred on a weekend

To see the distribution of the target variable, I first created a pie chart. Just 15.5% of visitors made a purchase, compared to 84.5% who did not.

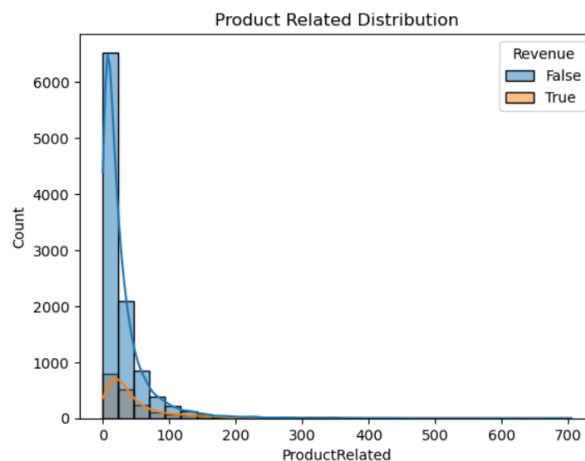


To view the outliers in the numerical fields, I then created boxplots. I chose not to eliminate any outliers even though the boxplots display outliers because there are too many points listed outside of the box.

I then plotted a few variables' counts against the desired variable.

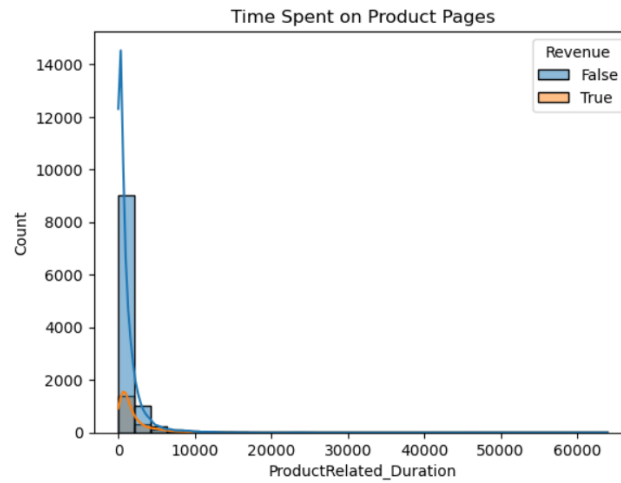
Number of product-related pages viewed Vs. Revenue

Since both have skewed lines on the bar chart, it is difficult to determine whether the quantity of product-related pages visited has any bearing on revenue.

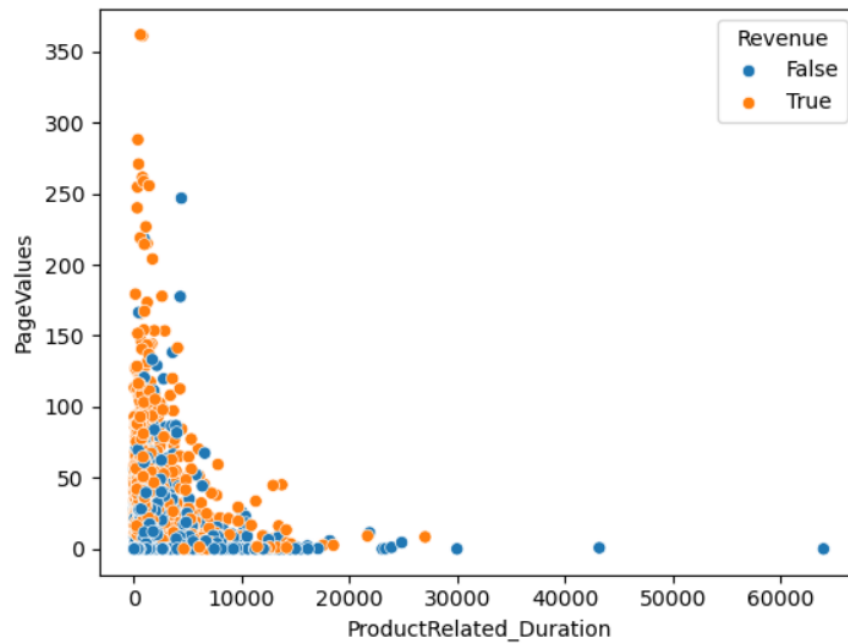


Time spent on product pages

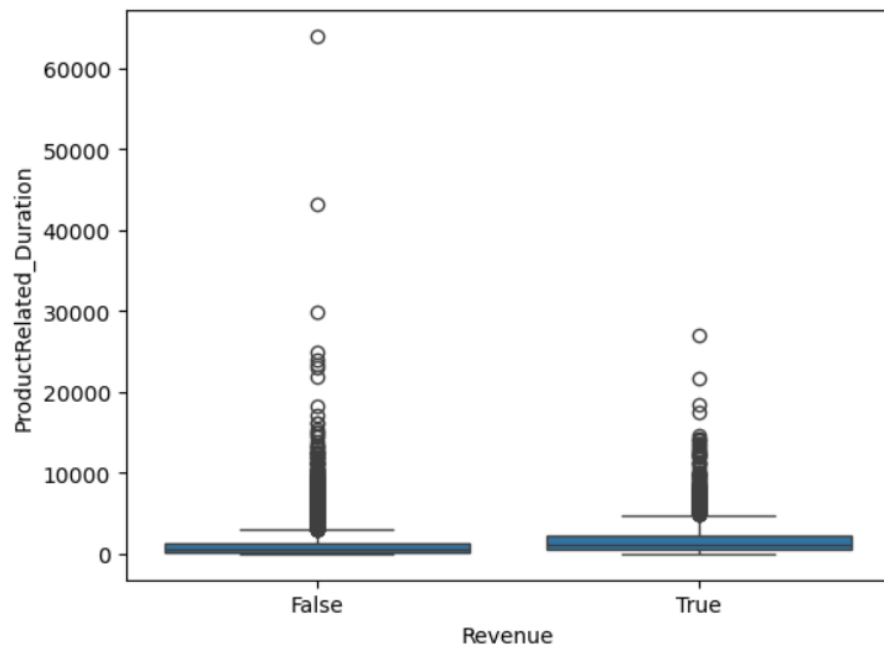
This is also unclear because it displays comparable patterns for both True and False Revenue.



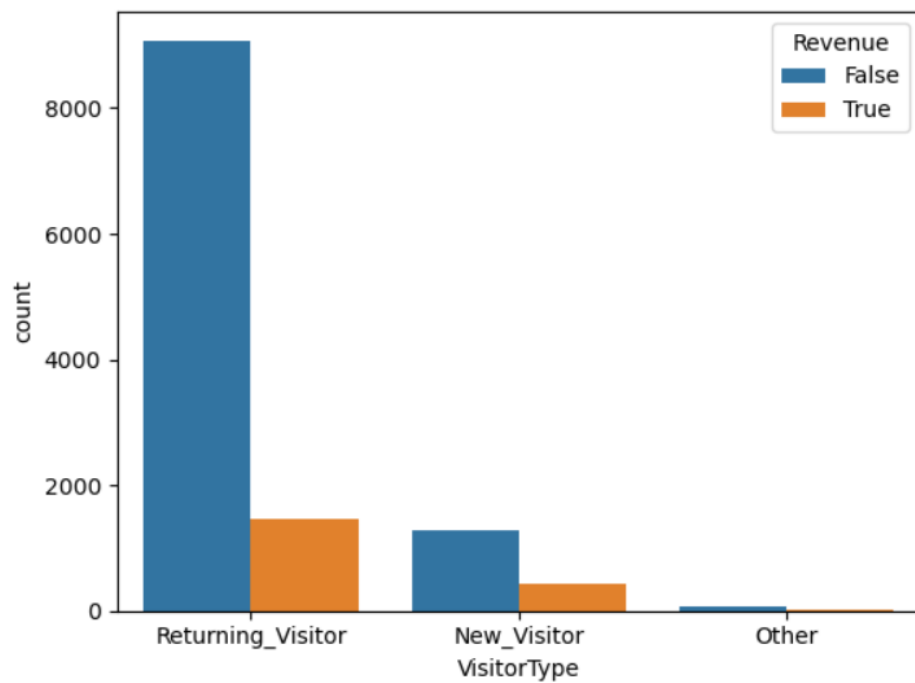
The scatter figure that follows demonstrates that when less time is spent on product-related pages, the page value has increased. Page value, as previously mentioned, is the page's worth prior to a transaction.



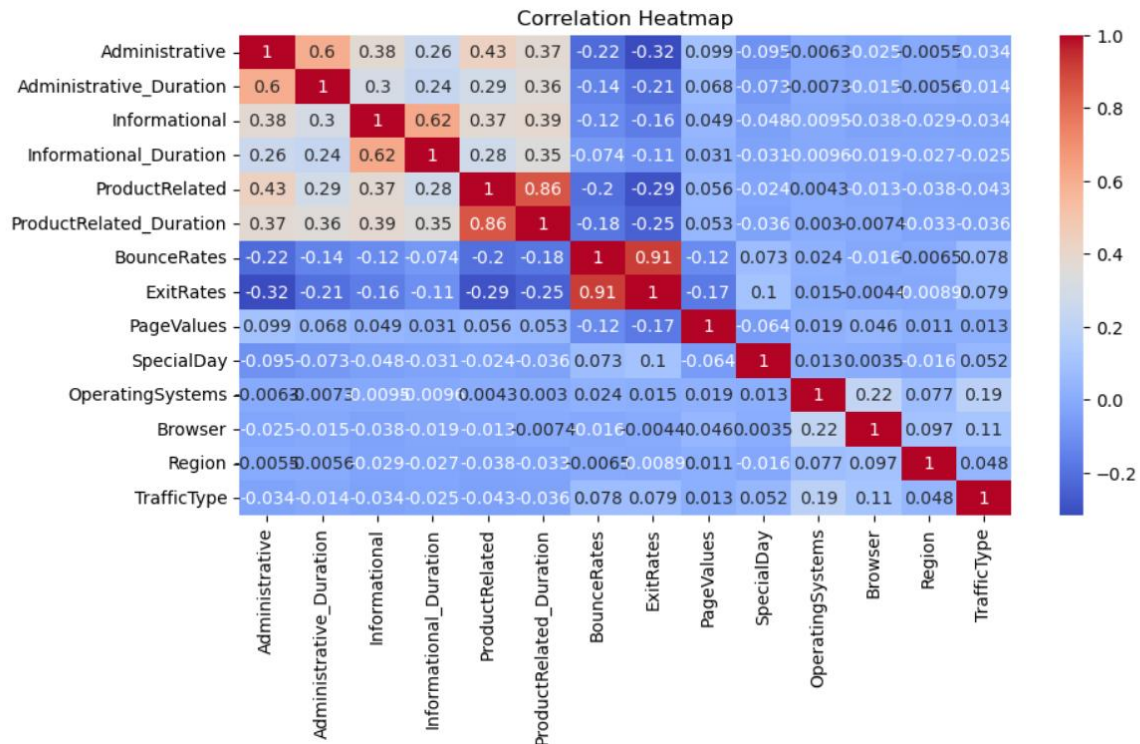
Based on the plots below, we could conclude that high-intention visitors spend less time on product-related pages.



Although there are more repeat visitors, their purchases are significantly lower than those of new visitors.



Finally, I plotted the **correlation matrix**



- There is a very substantial positive correlation (0.86) between ProductRelated and ProductRelated_Duration. This implies that consumers who spend more time on product-related pages also prefer to visit such pages more frequently.
- There is a very substantial positive connection (0.91) between BounceRates and ExitRates. In other words, individuals who quickly leave a page are also very likely to leave the website completely.
- I chose to retain all of these columns for model development even though these variables may have overlapping data because of their high multicollinearity.
- There is a moderately positive connection (0.60) between Administrative and Administrative_Duration. Consequently, consumers may spend more time on administrative sites if they visit them more frequently.
- There is a positive connection (0.62) between Informational and Informational_Duration. People who visit more informational pages also read them for longer.
- There are negative associations between ProductRelated and ProductRelated_Duration and:
 - Bounce Rates
 - Exit Rates
- It's possible that customers who actively peruse products are less likely to abandon the website right away.
- The majority of other traits have weak connections with the following variables:

- Operating Systems
 - The browser
 - Area
 - Type of Traffic
- As a result, neither geographic nor technical factors significantly affect browsing behaviour on their own.
- The majority of variables have rather weak direct connections with PageValues.
- This could imply that a single behavioural parameter may not be the only factor influencing purchase intent.
- It's crucial to remember that a weak association equals a useless feature. But for the initial iteration of my ML pipeline, I am maintaining all of these columns. Later, I will change the settings to see whether the results are improving.

6. Model Development

The project was designed as a binary classification issue to forecast the likelihood that an online customer would finish a transaction during a session. To find the best method for forecasting purchase intent, several machine learning models were put into practice and contrasted.

Models Used

Logistic Regression (Baseline Model)

Since logistic regression is easy to understand and works well for binary classification problems, it was chosen as the first baseline model. It offered a solid basis for contrasting more sophisticated models.

Random Forest

In order to capture more intricate behavioural interactions within the dataset, Random Forest was included as an ensemble learning model. Random Forest may simulate non-linear interactions between features like PageValues, BounceRates, ProductRelated pages, and session duration, in contrast to Logistic Regression.

Tuned Random Forest (GridSearchCV)

GridSearchCV was used to create a tailored version of Random Forest in order to maximise model performance through hyperparameter tuning. To determine which setup yielded the best predictive performance, many parameter combinations were assessed.

Why Tree-Based Models Performed Better

Non-linear patterns and feature interactions are frequently found in user behaviour data in e-commerce settings. For instance, a user may be more likely to make a purchase if they spend a lot of time on product pages and have low bounce rates. When it comes to capturing these intricate linkages, tree-based models like Random Forest outperform linear models like Logistic Regression.

Furthermore, Random Forest models are better suitable for behavioural analysis tasks because they are more resilient to feature interactions, outliers, and fluctuating feature relevance.

7. Model Evaluation

The machine learning models' performance was evaluated using a number of evaluation indicators.

Metrics Used

- Accuracy
- ROC-AUC Score
- Classification Report

Accuracy

The percentage of accurately predicted observations among all predictions is known as accuracy. Accuracy can be deceptive when dealing with unbalanced datasets, even though it offers a broad perspective of model performance.

ROC-AUC Score

The main evaluation metric was the Receiver Operating Characteristic – Area Under Curve (ROC-AUC) score. This score assesses how well the model can differentiate between users who make purchases and those who don't across a range of classification thresholds.

ROC-AUC was prioritized because the dataset contains class imbalance, where non-purchasing sessions significantly outnumber purchasing sessions. Furthermore, ROC-AUC is more appropriate than accuracy alone because the project uses probability-based predictions for consumer segmentation.

Classification Report

The classification report was used to evaluate precision, recall, and F1-score for both classes. These metrics provided deeper insight into how effectively the model identified purchasing customers while minimizing false predictions.

Following are the results of the training models.

Logistic Regression:

```
--- Logistic Regression ---
Accuracy: 0.8519870235198702
ROC-AUC: 0.896064254203053

Classification Report:
              precision    recall  f1-score   support

     0           0.95       0.87       0.91       2084
     1           0.52       0.75       0.61        382

 accuracy          0.85          2466
 macro avg         0.73          2466
 weighted avg      0.88          2466

Confusion Matrix:
[[1814  270]
 [  95 287]]
```

- 85% of the predictions were correct
- 89.6% of the times, the model has separated buyers from non-buyers
- This model has predicted “buyer” and it is correct 52% of the times and “non-buyers”, 95%.
- The model successfully finds 75% of actual buyers.
- A reasonable F1-score
- True negatives – 1814
- False positives – 270
- False negatives – 95
- True positives – 287

Random Forest:

```
--- Tuned Random Forest ---
Accuracy: 0.8868613138686131
ROC-AUC: 0.9266940338254062

Classification Report:
              precision    recall  f1-score   support

     0           0.95       0.92       0.93       2084
     1           0.61       0.73       0.67        382

 accuracy          0.89          2466
 macro avg         0.78          2466
 weighted avg      0.90          2466

Confusion Matrix:
[[1910  174]
 [ 105 277]]
```

- 88.6% of the predictions were correct
- 92.6% of the times, the model has separated buyers from non-buyers
- This model has predicted “buyer” and it is correct 61% of the times and “non-buyers”, 95%.
- The model successfully finds 73% of actual buyers.
- A reasonable F1-score
- True negatives – 1910
- False positives – 174
- False negatives – 105
- True positives - 277

Tuned Random Forest:

```

--- Random Forest ---
Accuracy: 0.8734793187347932
ROC-AUC: 0.9240121695088984

Classification Report:
              precision    recall  f1-score   support

     0           0.95       0.90       0.92       2084
     1           0.57       0.75       0.65        382

   accuracy          0.87          2466
  macro avg           0.76          2466
 weighted avg           0.89          2466

Confusion Matrix:
[[1867  217]
 [  95 287]]

```

- 87.3% of the predictions were correct
- 92.4% of the times, the model has separated buyers from non-buyers
- This model has predicted “buyer” and it is correct 57% of the times and “non-buyers”, 95%.
- The model successfully finds 75% of actual buyers.
- A reasonable F1-score
- True negatives – 1867
- False positives – 217
- False negatives – 95
- True positives - 287

8. Hyperparameter Tuning

To improve the performance of the Random Forest model, hyperparameter tuning was conducted using GridSearchCV.

GridSearchCV

Using cross-validation, GridSearchCV methodically assesses various hyperparameter combinations and chooses the configuration that delivers the highest evaluation score.

Parameters Tuned

The following Random Forest hyperparameters were optimized:

- `n_estimators` – Number of trees in the forest
- `max_depth` – Maximum depth of each decision tree
- `min_samples_split` – Minimum samples required to split a node
- `min_samples_leaf` – Minimum samples required at a leaf node

Benefits of Hyperparameter Tuning

Hyperparameter tuning improves model performance by identifying the most effective parameter configuration for the dataset. This procedure enhances generalisation, lowers overfitting, and raises prediction accuracy. The final model for segmentation and prediction tasks was the tweaked Random Forest model, which outperformed the baseline models.

9. Prediction Pipeline

A robust machine learning pipeline architecture was used in the project's design to increase reusability, scalability, and maintainability.

Pipeline Structure

preprocessing.py → train.py → predict.py

preprocessing.py

This module handles data preprocessing tasks such as:

- Missing value handling
- Encoding categorical variables
- Feature preparation
- Column consistency

The preprocessing module is designed to support both training data and newly uploaded datasets during inference.

`train.py`

This module performs:

- Model training
- Model comparison
- Hyperparameter tuning
- Model evaluation
- Model saving

The best-performing model is automatically selected and saved for future predictions.

`predict.py`

This module loads the saved model and generates predictions for new datasets. It also calculates purchase probabilities and assigns behavioral segments to users.

Key Features

Saved Model

The trained model is saved as:

`purchase_model.pkl`

This allows the model to be reused without retraining.

Feature Consistency

The project saves feature column information using:

`feature_columns.pkl`

This ensures that uploaded datasets maintain consistent feature ordering and structure during prediction.

Probability Predictions

Predictions are generated using:

`predict_proba()`

This enables probability-based customer segmentation and behavioral analysis.

10. Behavior Segmentation

The research not only forecasts sales but also divides users into behavioural groups according to their browsing habits and anticipated likelihood of making a purchase.

Segments Used

High Intent Buyer

Users with very high purchase probability and strong engagement patterns.

Potential Buyer

Users showing moderate purchase intent and meaningful browsing behavior.

Engaged Browser

Users who actively browse products and spend significant time on the website but may not currently show strong purchasing probability.

At-Risk Visitor

Users with high bounce rates or exit rates, indicating weak engagement and potential drop-off behavior.

Low Intent Visitor

Users with low engagement and low purchase probability.

Importance of Segmentation

Behavior segmentation transforms raw predictions into actionable business insights. Instead of simply predicting purchases, segmentation allows businesses to:

- Identify high-value customers
- Improve personalization
- Optimize marketing strategies
- Detect customer drop-off patterns
- Improve conversion rates

This makes the project more practical for real-world e-commerce analytics.

11. Dashboard

An interactive dashboard was developed to visualize model outputs, customer segments, and behavioral insights.

Framework Used

The dashboard was implemented using Dash and Plotly visualizations.

Dashboard Features

- **KPI Metrics**

The dashboard displays key metrics such as:

- Total visitors
- Conversion rate
- Average purchase probability
- Number of buyers

- **Segment Distribution**

Visualizations show how users are distributed across behavioral segments.

- **Behavior Analysis**

Interactive charts analyze:

- Bounce rates
- Exit rates
- Product engagement
- Page values

- **Interactive CSV Upload**

Users can upload new CSV datasets directly through the dashboard interface.

- **Real-Time Predictions**

Uploaded datasets are processed through the trained machine learning pipeline to generate live predictions and behavioral segmentation.

I've experimented with few Dashboard types

1. Created a Dashboard with Streamlit – difficulty level easy
2. Created a Dashboard with Dash by Plotly -Dash is like building a web page with Python
3. Created a Dashboard with Flask + HTML/CSS – This is like a web project
4. Finally developed the Dash dashboard to handle CSV upload

12. System Workflow

The complete workflow of the project is illustrated below:

- Raw Data
- Preprocessing
- Model Training
- Model Evaluation
- Hyperparameter Tuning
- Model Saving
- Prediction
- Behavior Segmentation
- Dashboard Visualization
- User CSV Upload
- Live Predictions

This workflow demonstrates a complete end-to-end machine learning pipeline capable of handling both training and inference tasks.

13. Results & Insights

The tuned Random Forest model achieved the strongest overall performance based on ROC-AUC evaluation.

Key Findings

- Returning visitors showed higher purchase intent compared to new visitors
- Higher PageValues strongly correlated with purchase probability
- High BounceRates and ExitRates negatively impacted conversions
- Product-related engagement strongly influenced purchasing behavior

Segment Insights

- The segmentation process revealed distinct user groups ranging from highly engaged buyers to low-intent visitors. These insights can assist businesses in customer targeting, personalization, and conversion optimization.

14. Limitations

Despite strong performance, the project contains several limitations.

- **Dataset Size Limitations:** The dataset size is relatively limited compared to large-scale commercial e-commerce systems.
- **Static Model:** The model is trained offline and does not currently support automated real-time retraining.
- **Assumption of Similar Feature Distribution:** The prediction pipeline assumes that newly uploaded datasets follow a similar feature structure and behavioral distribution as the original training dataset.

15. Future Improvements

Several improvements can further enhance the project.

Advanced Models

Implementing models such as XGBoost or LightGBM may improve predictive performance.

Cloud Deployment

The application can be deployed as a cloud-based web application for broader accessibility.

Real-Time Data Pipelines

Integrating streaming or real-time data pipelines would improve scalability and production readiness.

Advanced Segmentation

Customer segmentation can be improved using clustering techniques such as K-Means or DBSCAN for more data-driven grouping.

16. How to Run the Project

Step 1 – Activate Environment

```
conda activate env
```

Step 2 – Install Dependencies

```
pip install -r requirements.txt
```

Step 3 – Train the Model

```
python src/train.py
```

Step 4 – Launch Dashboard made with Streamlit

```
streamlit run app/dashboard.py
```

Step 5 – Launch Dashboard made with Dash by Plotly

```
python app/dashboard_dash.py
```

Step 6 – Launch Dashboard made with Flask + HTML/CSS

```
python app/app.py
```

Step 7 – Launch Dashboard made with Dash to upload CSV files

```
python app/dashboard_dash_with_upload.py
```

17. Conclusion

This project successfully developed an end-to-end machine learning pipeline for analyzing online shopper behavior and predicting purchase intent.

The project combined:

- Data preprocessing
- Machine learning model development
- Hyperparameter tuning
- Customer behavior segmentation
- Interactive dashboard visualization

In addition to generating predictions, the system provides meaningful behavioral insights that can support marketing optimization, customer targeting, and personalization strategies in e-commerce environments.

Technically, the project demonstrates skills in:

- Machine learning pipeline design
- Data preprocessing
- Model evaluation
- Hyperparameter tuning
- Dashboard development
- Real-time prediction workflows

Overall, the project highlights how behavioral analytics and machine learning can be integrated to support data-driven decision-making in modern e-commerce systems.